FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Main)-2023

TEST DATE: 28-03-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C

Sol. The distance covered in 2nd second is given by

$$S_2 = v_0 + \frac{a}{2}(2\times 2 - 1)$$

The distance covered in the 4th second is given by

$$S_4 = v_0 + \frac{a}{2}(2\times 4 - 1)$$

$$v_0 = \frac{a}{2} = \frac{3}{2} = 1.5\ ms^{-1}$$

2. A

Sol. $I = \frac{6}{1+2} = 2A$

D_1 is forward biased and D_2 is reversed biased

3. C

Sol. $T_{1/2} = \frac{0.693}{\lambda}$

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{30} = \frac{0.0693}{3} = 0.0231 \text{ per day} = 231 \times 10^{-4} \text{ per day}$$

4. C

Sol. $\lambda = \frac{h}{\sqrt{2Em}} = \frac{6.6\times 10^{-34}}{\sqrt{2\times 0.08\times 1.6\times 10^{-19}\times 1.67\times 10^{-27}}}$

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5. B

Sol. Since Q is the first maximum after P, AQ – BQ = λ

BQ = 2.4 m, AB = 1.0 m, ABQ = 90°

Hence AQ $= \sqrt{BQ^2 + AB^2} = \sqrt{(2.4)^2 + (1.0)^2} = 2.6$ m

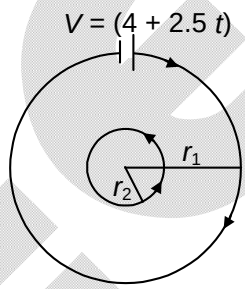
Also $\lambda = 2.60 - 2.40 = 0.20$ m

Hence wave speed, C $= 340 = f \cdot \lambda = f \cdot (0.20)$

Hence frequency, $f = \dfrac{340}{0.20} = 1700$ Hz

6. D

Sol. $r_1 = 1.0$ m, $r_2 = 10^{-1}$ m



Resistance of outer loop $= 2\pi \times 10^{-4}\ \Omega$

Resistance of inner loop $= 0.2\pi \times 10^{-4}\ \Omega$

Current in outer loop $= \dfrac{V}{R} = \dfrac{(4 + 2.5\ t)}{2\pi \times 10^{-4}}$ A

or $i_0 = \left[\left(\dfrac{2}{\pi}\right) \times 10^4 + \left(\dfrac{1.25}{\pi}\right) \times 10^4 \times t\right]$ A

Magnetic field produced at the common centre (see figure)

$B = \dfrac{\mu_0 i}{2r_1} = \left(\dfrac{\mu_0 i}{2}\right)$ T

or $B = \dfrac{4\pi}{2} \times 10^{-7} \times \dfrac{[(2 + 1.25\ t)\ 10^4]}{\pi} = 2 \times 10^{-3}\ (2 + 1.25\ t)$ T

Hence, flux linked with the inner loop,

$\phi = BA = 2 \times 10^{-3}\ (2 + 1.25\ t) \times \pi\ (0.1)^2 = 2\pi \times 10^{-5}\ (2 + 1.25\ t)$ Wb.

Hence, e.m.f. induced in smaller loop $= \varepsilon = -\dfrac{d\varphi}{dt} = -2\pi \times 10^{-5} \times 1.25$

$= -2.5\ \pi \times 10^{-5}$ V.

The negative sign indicates that the induced e.m.f. (or current) is opposite to applied e.m.f. (or current).

Hence, the current induced in the inner (smaller) loop is

$= \dfrac{|\varepsilon|}{R} = \dfrac{2.5\pi \times 10^{-5}\ V}{(0.2\pi \times 10^{-4})\ \Omega} = 1.25$ A.

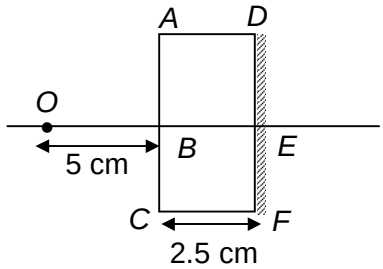
7. D

Sol. Let I_1, I_2 and I_3 be the images formed by

(i) refraction from ABC

(ii) reflection from DEF and

(iii) again refraction from ABC



Then $BI_1 = (5)\ \mu_g = (5)\ (1.5) = 7.5$ cm

Now $EI_1 = (7.5 + 2.5) = 10$ cm

$\therefore$ $EI_2 = 10$ cm behind the mirror

Now $BI_2 = (10 + 2.5) = 12.5$ cm

$\therefore$ $BI_3 = \dfrac{12.5}{\mu_g} = \dfrac{12.5}{1.5} = 8.33$ cm

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8. B

Sol. Since AB, CD, EF have equal resistances each carry 10 A current, force on AB per unit length due to CD

$$= \frac{\mu_0}{2\pi} \times \frac{10 \times 10}{1 \times 10^{-2}} = \frac{\mu_0}{2\pi} \times 10^4$$

Force on AB per unit length due to EF

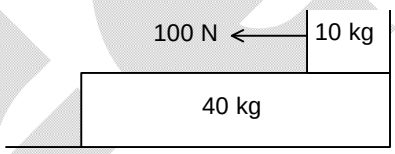
$$= \frac{\mu_0}{2\pi} \times \frac{10 \times 10}{2 \times 10^{-2}} = \frac{\mu_0}{4\pi} \times 10^4$$

Total force downwards $= \frac{\mu_0}{4\pi} \times 10^4 \times 3 = 3 \times 10^{-3}$ N/m

Force on CD due to AB and EF are equal and opposite and the net force is zero.

9. A

Sol. For a force of 100 N on 10 kg block, relative motion will take place.



$\therefore$ The frictional force between 10 kg block and 40 kg block,

$f = \mu mg$

$= 0.4 \times 10 \times 9.8$ N

The acceleration of the slab of 40 kg is

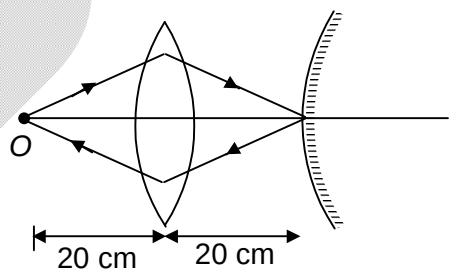
$$a = \frac{0.4 \times 10 \times 9.8}{40} = 0.98 \text{ m/s}^2$$

10. C

Sol. Basic concept

11. C

Sol. Object is placed at a distance of 2f from the lens (f = focal length of lens) i.e., the image formed by the lens will be at a distance of 2f or 20 cm from the lens. So, if the concave mirror is placed in this position, the first image will be formed at its pole and it will reflect all the rays symmetrically to other side as shown below and the final image will coincide with the object.



12. D

Sol. Since field is uniform, the net flux for the closed surface is zero.

13. A

Sol. With the help of FBD of m, we can write

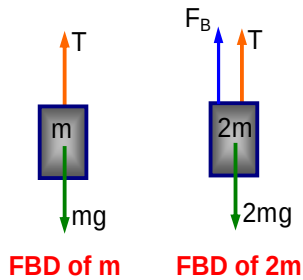
$T = mg$

With the help of FBD of 2m, we can write

$T + F_B = 2mg$

$$\Rightarrow mg + \frac{2}{3}\left(\frac{2m}{\rho}\right)\rho_0 g = 2mg$$

$$\Rightarrow \frac{4\rho_0}{3\rho} = 1 \Rightarrow \rho = \frac{4}{3}\rho_0 = 1.339 \text{ g/cm}^3$$



FBD of m FBD of 2m

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14. B

Sol. Heat required at constant pressure to raise the temperature by ΔT : $Q_1 = nC_p \Delta T$

Heat required at constant volume for the same rise in temperature: $Q_2 = nC_v \Delta T$

$$\therefore \quad \frac{Q_2}{Q_1} = \frac{nC_v\Delta T}{nC_p\Delta T} = \frac{C_v}{C_p} = \frac{1}{\gamma} = \frac{5}{7}$$

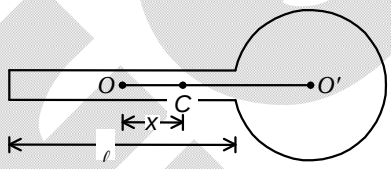
$$\Rightarrow \quad Q_2 = \frac{5}{7} \times Q_1 = \frac{5}{7} \times 70 = \mathbf{50\ J}.$$

15. A

Sol. If distance between the two stars is 3 l then angular momentum of mass m about common centre of mass = m $(2l)^2$ ω and angular momentum of mass 2m about common centre of mass = (2m) $l^2\omega$.

16. D

Sol. Let O be the centre of mass of the handle and O′ that of the disc. Let x be the distance of C.M of the system C from O.



Then $mx = m\left(\frac{\ell}{2} + r - x\right)$

$$x = \frac{\ell + 2r}{4} = OC = O'C$$

$$\text{required moment of inertia} = \left[\frac{m\ell^2}{12} + m\left(\frac{\ell+2r}{4}\right)^2 + \left[\frac{mr^2}{2} + m\left(\frac{\ell+2r}{4}\right)^2\right]\right]$$

$$= \frac{m\ell^2}{12} + \frac{mr^2}{2} + \frac{m}{8}(\ell + 2r)^2$$

17. D

Sol. Extension: $\Delta l = \frac{Fl}{AY}$

For the bar to remain horizontal, $\Delta l_1 = \Delta l_2$

$\Rightarrow A_1Y_1 = A_2Y_2$ ($\because$ F and ℓ are common)

$\Rightarrow d_1^2Y_1 = d_2^2Y_2$ ($\because A \propto d$)

$$\frac{d_1}{d_2} = \sqrt{\frac{Y_2}{Y_1}}$$

$\therefore$ (D)

18. A

Sol. Before entering the water the velocity of ball is $\sqrt{2gh}$. If after entering the water this velocity does not change then this value should be equal to the terminal velocity. Therefore,

$$\sqrt{2gh} = \frac{2}{9}\frac{r^2(\rho - \sigma)g}{\eta}$$

$$h = 1.65 \times 10^3\ m$$

19. C

Sol. $2 = (i_1 + i_2)\ 0.5 + 1i_2$

$i_1 = i_2$

$2 = 2i_1\ (0.5) + i_1$

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$= 2i_1$

i_1 $= 1$ A, $i_1 = 1$ A

$\therefore$ the current through 0.5 Ω, resistor $i = i_1 + i_1 = 2$ A

Maximum power developed across $R = i^2R = 2^2 (0.5) = 2$ W

20. A

Sol. Path difference $\Delta x = \frac{yd}{D}$

here $y = \frac{d}{2} = \frac{5\lambda}{2}$ (as $d = 5\lambda$) and $D = 10\ d = 50\ \lambda$

so $\Delta x = \left(\frac{5\lambda}{2}\right)\left(\frac{5\lambda}{50\lambda}\right) = \frac{\lambda}{4}$

corresponding phase difference will be

$\phi = \left(\frac{2\pi}{\lambda}\right)(\Delta x) = \left(\frac{2\pi}{\lambda}\right)\left(\frac{\lambda}{4}\right) = \frac{\pi}{2}$

or $\frac{\varphi}{2} = \frac{\pi}{4}$

$\therefore$ $I = I_0 \cos^2\left(\frac{\varphi}{2}\right)$

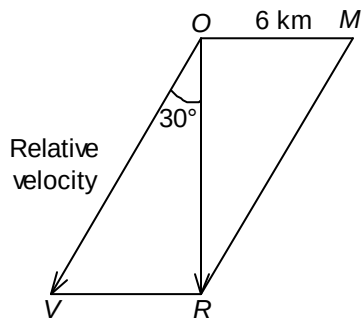
$I_0 \cos^2\left(\frac{\pi}{4}\right) = \frac{I_0}{2}$

SECTION – B

21. 9

Sol. Velocity of rain = Velocity of man + Relative velocity of rain

OR gives the actual velocity.



$\tan 30° = \frac{VR}{OR} = \frac{1}{\sqrt{3}} = \frac{6}{OR}$

or, $OR = \mathbf{6\sqrt{3}}$ km/hr

22. 1215

Sol. Balmer series

$\frac{1}{\lambda_1} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5R}{36}$

Lyman series

$\frac{1}{\lambda_2} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3R}{4}$

$\frac{\lambda_2}{\lambda_1} = \frac{4}{3R} \times \frac{5R}{36} = \frac{20}{108} = \frac{5}{27}$

$\lambda_2 = \frac{5}{27} \times \lambda_1 = \frac{5}{27} \times 6563 = 1215$ Å

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23. 1000

Sol. The velocity of sound in seawater is $v = \sqrt{\frac{E}{\rho}}$

$$v = \sqrt{\frac{E}{\rho}} = \sqrt{\frac{220 \times 10^6}{1.1 \times 10^3}} = 100\sqrt{20}\ \text{m/s}$$

$$2y = v_W \Delta t$$

$$y = \frac{v_w \Delta t}{2} = \frac{100\sqrt{20} \times \sqrt{20}}{2}$$

The depth of the well is therefore = 1000 m

24. 1

Sol. The instantaneous magnetic flux through the loop is

$$\Phi_B = BA\cos\theta$$

Since, $\theta = \omega t$, therefore, $\Phi_B = BA\cos\omega t$

From Faraday's law, equation

$$e = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\left[BA\cos\omega t\right] \text{ or } e = BA\omega\sin\omega t$$

The induced current is $I = \frac{e}{R} = \left(\frac{BA\omega}{R}\right)\sin\omega t$

The amplitude of induced current (in mA) $= \frac{BA\omega}{R} = 1$

25. 50

Sol. The impedance of the circuit is given by

$$Z = \sqrt{(R^2 + \omega^2 L^2)} = \sqrt{[R^2 + (2\pi fL)^2]}$$

$$= \sqrt{[(12)^2 + \{2 \times 3.14 \times 50 \times (0.05/3.14)\}^2]} = \sqrt{(144 + 25)} = 13 \text{ ohm}$$

Current in the circuit $i = E/Z = \frac{130}{13} = 10$ amp.

Inductive reactance of coil $X_L = \omega L = 2\pi fL$

$\therefore X_L = 2\pi \times 50 \times \left(\frac{0.05}{\pi}\right) = 5$ ohm.

Potential difference across inductance

$V_L = i \times X_L = 10 \times 5 = 50$ volt.

26. 7

Sol. For maximum intensity on the screen

$d\sin\theta = n\lambda$

$$\text{or } \sin\theta = \frac{n\lambda}{d}$$

$$= \frac{(n)(2000)}{(7000)} = \frac{n}{3.5}$$

Since, $\sin\theta \le 1$

$\therefore$ n = 0, 1, 2, 3 only.

Thus, only seven maxima's can be obtained on both sides of the screen.

27. 14

Sol. $v = \sqrt{\mu gr} = \sqrt{0.2 \times 100 \times 9.8} = 14$ m/s

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28. 7

Sol. The current will be conducted radially outwards from the inner conductor to the outer conductor. The area of cross-section for the conduction of the current is therefore the area of an elementary cylindrical shell which varies with radius. The length of the conducting shell is measured radially from radius a to radius b.

Consider an elementary cylindrical shell of radius r and thickness dr. Its area of cross-section (normal to flow of current) = $(2\pi r\ell)$ and its length = dr.

Hence the resistance of the elementary cylindrical shell of the medium is

$$dR = \frac{\rho dr}{2\pi r\ell} = \frac{\rho}{2\pi\ell}\left[\frac{dr}{r}\right]$$

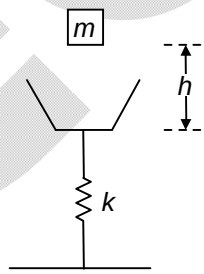
The resistance of the medium is obtained by integrating for r from a to b.

Hence required resistance

$$R = \frac{\rho}{2\pi\ell}\int_a^b \frac{dr}{r} = \frac{\rho}{2\pi\ell}\left[\log_e r\right]_a^b \left(\frac{\rho}{2\pi\ell}\right)\log_e \frac{b}{a} = 7\ \Omega$$

29. 11

Sol. Suppose by falling down through a height h, the mass m compresses the spring balance by a length x.



The P.E. lost by the mass = mg (h+ x)

This is stored up as energy of the spring by compression = $\frac{1}{2}kx^2$

$$\therefore \quad mg(h + x) = \frac{1}{2}kx^2 \text{ or } \frac{1}{2}kx^2 - mgx - mgh = 0$$

or $$x^2 - \frac{2mgx}{k} - \frac{2mgh}{k} = 0$$

Solving this quadratic equation, we get

$$x = \frac{\frac{2mg}{k} \pm \sqrt{\left(\frac{2mg}{k}\right)^2 + \left(\frac{8mgh}{k}\right)}}{2} = \frac{mg}{k} \pm \frac{mg}{k}\sqrt{1 + \frac{2kh}{mg}}$$

In the equilibrium position, the spring will be compressed through the distance mg/k and hence the amplitude of oscillation is

$$A = \frac{mg}{k}\sqrt{1 + \frac{2kh}{mg}}$$

Energy of oscillation = $\frac{1}{2}kA^2 = \frac{1}{2}k\left(\frac{mg}{k}\right)^2\left(1 + \frac{2kh}{mg}\right)$

$$= mgh + \frac{(mg)^2}{2k} = 11\ J$$

30. 5

Sol. $(18 - 11) = \frac{Q}{2\mu F} + \frac{Q}{5\mu F} = \frac{7Q}{10\mu F}$

$Q = 10\mu C$

$V_A = \frac{Q}{C_A} = \frac{10\mu C}{2\mu F} = 5\ V$

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Chemistry

PART – B

SECTION – A

31. C

Sol. (A) Negatively charged sol is produced.

(B) No colloid formation occurs in cold water due to no hydrolysis of $FeCl_3$ in cold water.

(D) Gold sol is a lyophobic sol.

32. A

Sol. $\Delta\ell = \pm 1$ only; for allowed transition.

33. D

Sol. (A) NO_3^- & CO_3^{2-} both are trigonal planar and isoelectronic as well.

(B) $CO_2, \&\overset{\oplus}{N}O_2$ both are linear, having equal electron.

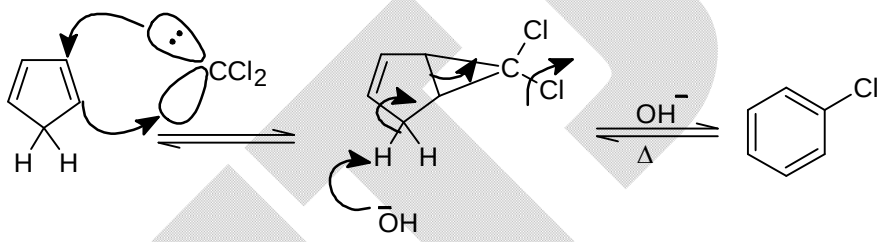
(C) XeF_2 & IF_2^- both are isoelectronic and linear in shape, with trigonal bipyramidal geometry.

34. B

Sol. The fraction of molecules with too low or too high speeds is very small. And on increasing temperature the fraction of molecules attaining most probable speed decreases.

35. D

Sol.



36. A

Sol. $Hg(NO_3)_2 \xrightarrow{\Delta} Hg(\ell) + 2NO_2 + O_2$;

37. B

Sol. $Fe^{3+} + K_4\left[Fe(CN)_6\right] \longrightarrow Fe_4\left[Fe(CN)_6\right]_3 \downarrow + K^+$

(Prussian blue)

38. A

Sol. $\left[Cr(H_2O)_6\right]^{3+}$ - Violet;

$\left[Cr(H_2O)_4Cl_2\right]^+$ - Green;

CrO_4^{2-} - Yellow;

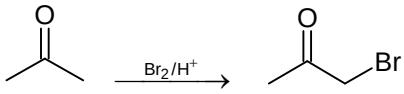
$\left[Fe(CN)_6\right]^{4-}$ - Yellow;

39. D

Sol. Carboxylic acids are generally more acidic than alcohols, oximes and ammonium ions.

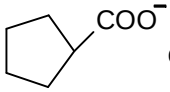
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40. C

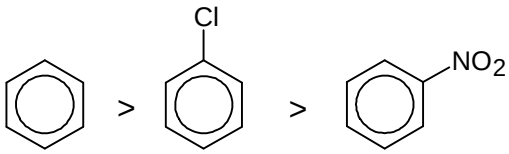
Sol. (A) 

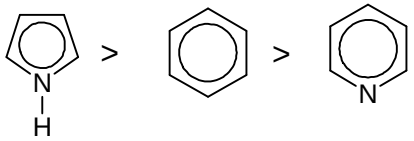
(B) No reaction.

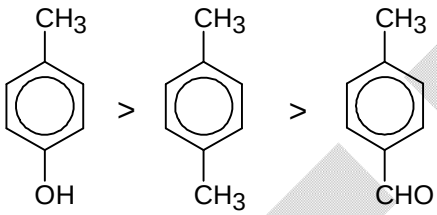
(C) $HCBr_3$ is formed along with CH_3COO^-.

(D)  can be the product instead of chloroform.

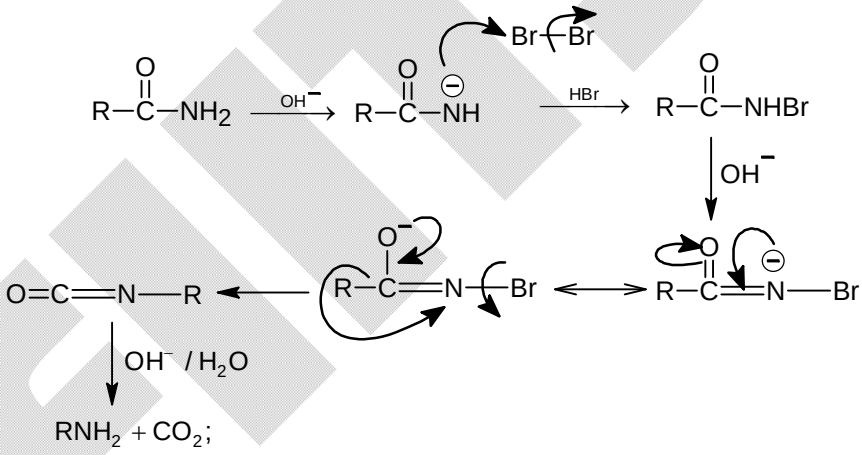
41. B

Sol. (A) 

(C) 

(D) 

42. D

Sol. 

43. A

Sol. Due to higher overvoltage of production of $O_2(g)$, and low overvoltage of liberation of Cl_2 (g) is obtained at anode. pH of the following solution increases due to the following process at cathode:

$2H_2O + 2e^- \longrightarrow H_2 \uparrow + 2OH^-;$

44. B

Sol. $Ni \quad + \quad 4CO \xrightarrow{320\ K} Ni(CO)_4 \xrightarrow{420\ K} Ni + 4CO;$

$(Impure) \qquad\qquad (Pure)$

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45. D

Sol. In $\left[V(CO)_6\right]^-$ electron pair donation from metal to 'CO' is the best hence CO – bond order decreases and bond length increases to much extent.

46. B

Sol. Ranitidine is an antacid. Dichlofenac and Aspirin are achiral.

47. B

Sol. (A) In Down's process, fused NaCl, $CaCl_2$ mixture is electrolysed.
(C) In Nelson process, electrolysis of Brine takes place.
(D) In Gossage process, Na_2CO_3 and $Ca(OH)_2$ are raw materials.

48. C

Sol. (A) 1°-carbocation formation is not favourable under given conditions.
(B) Free radical process occurs in presence of peroxide, ROOR.
(D) S_N2 reaction occurs.

49. A

Sol. $AgCl(s) \underset{\rightleftharpoons}{\overset{H_2O}{\rightleftharpoons}} Ag^+(aq) + Cl^-(aq)$;

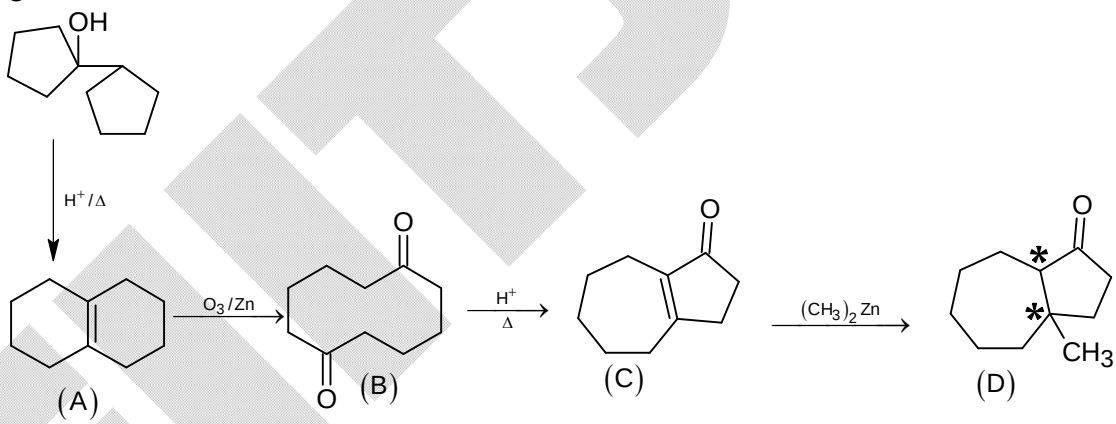
$2Na_2S_2O_3 \longrightarrow 4Na^+(aq) + 2S_2O_3^{2-}$;

$Ag^+ + 2S_2O_3^{2-} \underset{\rightleftharpoons}{\overset{H_2O}{\rightleftharpoons}} \left[Ag(S_2O_3)_2\right]^{3-}$;

As the number of ions in solution increasing, hence conductivity increases.

50. C

Sol.



SECTION – B

51. 4

Sol. At 50% equivalence point [Salt] = [Acid];
Hence, $pH = pK_a = 4$

52. 8

Sol. Q (heat) and W(work) are path functions while other 8 are state functions.

53. 8

Sol. If we consider a corner Cl^- ion then its third nearest neighbour are body centred Na^+ ions.

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54. 4

Sol. % B in product mixture $= \dfrac{2\times3\times10^{-3}}{2\times3\times10^{-3}+3\times4.8\times10^{-2}}\times100 = 4$

55. 7

Sol. $AgOH(s) \rightleftharpoons Ag^{+} + OH^{-}; \quad K_{sP}$

$S \quad S$

$\sim$

$Ag^{+} + 2NH_3 \rightleftharpoons \left[Ag(NH_3)_2\right]^{+}; \ K_f$

$S \quad 0.1$

$\sim \quad 0.1-2S \quad \approx S$

$AgOH(s) + 2NH_3 \rightleftharpoons \left[Ag(NH_3)_2\right]^{+} + OH^{-}; \qquad K_3 = K_{SP}\times K_f$

$0.1-2S \quad \simeq S \quad S$

$2\times10^{-8}\times1.7\times10^{+7} = \dfrac{S^2}{(0.1-2S)^2}$

$S = 2.69\times10^{-2} = \left[OH^{-}\right]$

For AgOH $\Rightarrow K_{SP} = \left[Ag^{+}\right]\left[OH^{-}\right] \Rightarrow \left[Ag^{+}\right] = \dfrac{2\times10^{-8}}{2.69\times10^{-2}} = 7.434\times10^{-7}$ M

56. 7

Sol. H_2O, I_3^-, XeF_4, Be_2H_4, SCl_2, XeF_2 and I_2Cl_6

57. 4

Sol. Fe, Pb, Zn, Sn.

58. 7

Sol. Except $Fe(OH)_3$ and Ag_2S.

59. 9

Sol. It is Carius method of estimation of Bromine.

$\%Br = \dfrac{80}{188}\times\dfrac{0.1692}{0.8}\times100 = 9$

60. 4

Sol. According to selection rules, the electronic transitions mentioned in a, b, c and e are permissible while others are forbidden.

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Mathematics

PART – C

SECTION – A

61. C

Sol. Subsets of A whose intersection with B is a null set will be 2^6 (which are subsets of {1, 3, 4, 6, 7, 9}) $\therefore$ required number of sets = $2^9 - 2^6 = 448$

62. A

Sol. $\left|z-4\sqrt{3}\right|+\left|z-k\sqrt{3}i\right|=\left|z-4\sqrt{3}\right|+\left|k\sqrt{3}i-z\right|$

$\geq \left|z-4\sqrt{3}+k\sqrt{3}i-z\right| \geq \sqrt{3k^2+48}$

63. B

Sol. $\dfrac{9!(x)^{a_1}(2y)^{a_2}(3xy)^{a_3}}{a_1!a_2!a_3!}$ where $a_1 + a_2 + a_3 = 9$

$= \dfrac{9!x^{a_1+a_3}y^{a_2+a_3}2^{a_2}3^{a_3}}{a_1!a_2!a_3!}$

$\Rightarrow$ $a_1 + a_3 = 6$, $a_2 + a_3 = 4$ $\therefore$ $a_1 = 5$, $a_2 = 3$, $a_3 = 1$

$\Rightarrow \dfrac{9!2^3 3}{5!3!1!} = k \times 2^6 \times 3^3 \Rightarrow k = 7$

64. B

Sol. $\dfrac{1}{15}\left(\dfrac{1}{(15-k)}-\dfrac{1}{(30-k)}+\dfrac{1}{(30-k)}+.....+\dfrac{1}{(300-k)}-\dfrac{1}{(315-k)}\right)=\dfrac{1}{155}$

$\Rightarrow \dfrac{1}{15}\left(\dfrac{300}{(15-k)(315-k)}\right)=\dfrac{1}{155} \Rightarrow k = 5, 325$

65. C

Sol. Centre : $\left(\dfrac{1}{2},-\dfrac{3}{2}\right)$ radius = $\sqrt{\dfrac{5}{2}}$

Let L_1 be $y = mx$

$BD^2 = AB^2 - AD^2 = \dfrac{5}{2}-\left(\dfrac{\left|\frac{1}{2}-\frac{3}{2}-1\right|}{\sqrt{2}}\right)^2$

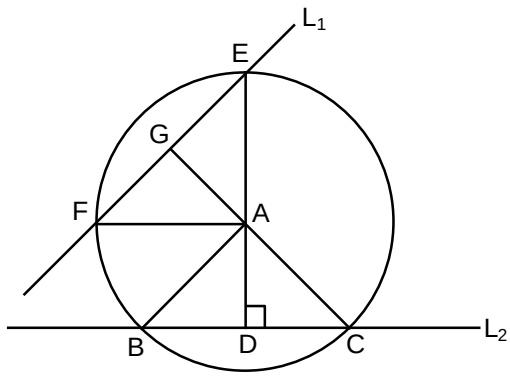
$AF^2 = FG^2 + AG^2$

$FG^2 = \dfrac{5}{2}-\left(\dfrac{\left|\frac{m}{2}+\frac{3}{2}\right|}{\sqrt{m^2+1}}\right)^2 \Rightarrow \dfrac{(m+3)}{2\sqrt{m^2+1}} = \sqrt{2}$

$m^2 + 9 + 6m = 8m^2 + 8$

$7m^2 - 6m - 1 = 0$

$m = 1, -\dfrac{1}{7}$.



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66. C

Sol. P(3cosθ, 4sinθ)

Equation of chord of contact xcosθ + ysinθ = 1

It passes through (1, 1) $\Rightarrow$ cosθ + sinθ = 1

67. D

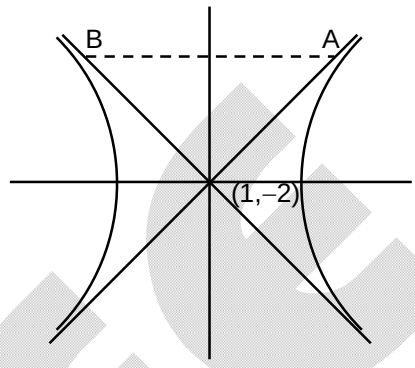
Sol. $(x-1)^2 - 4(y+2)^2 = 16$

Equation of asymptotes $(x-1)^2 - 4(y+2)^2 = 0$

$\Rightarrow$ x – 2y – 5 = 0 and x + 2y + 3 = 0

(α, 2) must lie between A and B

$\Rightarrow \alpha \in (-7, 9)$



68. C

Sol. Distance between vertex and directrix = $\left|\frac{3+4+3}{5}\right| = 2$

$\therefore$ length of L.R = 8

69. C

Sol. P(E) = P(X = 3) + P(X = 5) + P(X = 7) + P(X = 11)

= 0.07 + 0.08 + 0.20 + 0.05 = 0.4

P(F) = 0.12 + 0.05 + 0.07 + 0.1 +).08 + 0.03 = 0.45

$P(E \cap F)$ = P(X = 3) + P(X = 5) = 0.07 + 0.08 = 0.15

70. C

Sol. Let image of this point be (x_2, y_2, z_2)

$$\frac{x_2-(2\lambda+1)}{1} = \frac{y_2-(\lambda)}{-4} = \frac{z_2-(2\lambda-1)}{1} = \frac{-2(2\lambda+1-4\lambda+2\lambda-1-9)}{18}$$

$\Rightarrow (2\lambda + 2, \lambda - 4, 2\lambda)$

(C) satisfies it for $\lambda = 1$

71. A

Sol. $\left[(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}.\vec{b})\vec{c},(\vec{b}\cdot\vec{a})\vec{c}-(\vec{b}\cdot\vec{c})\vec{a},(\vec{c}\cdot\vec{a})\vec{b}-(\vec{c}\cdot\vec{b})\vec{a}\right]$

$= \left[4\vec{b}-\vec{c},\vec{c}+2\vec{a},4\vec{b}+2\vec{a}\right]$

$= \left(4\vec{b}-\vec{c}\right)\cdot\left((\vec{c}+2\vec{a})\times\left(4\vec{b}+2\vec{a}\right)\right)$

$= \left(4\vec{b}-\vec{c}\right)\cdot\left(4\vec{c}\times\vec{b}+2\vec{c}\times\vec{a}+8\left(\vec{a}\times\vec{b}\right)\right)$

$= 8\left[\vec{b}\,\vec{c}\,\vec{a}\right]-8\left[\vec{c}\,\vec{a}\,\vec{b}\right] = 0$

72. A

Sol. α + β + γ + δ = 0

$\therefore \quad -\frac{1}{\alpha}-\frac{1}{\beta}-\frac{1}{\gamma}-\frac{1}{\gamma} = \left(\frac{\alpha\beta\gamma+\beta\gamma\delta+\alpha\beta\delta+\alpha\gamma\delta}{\alpha\beta\gamma\delta}\right) = -\left(-\frac{2}{3}\right) = \frac{2}{3}$

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73. D

Sol. $941111 \rightarrow \frac{6!}{4!} = 30$

$661111 \rightarrow \frac{6!}{4!2!} = 15$

$236111 \rightarrow \frac{6!}{3!} = 120$

$232311 \rightarrow \frac{6!}{2!2!2!} = 90$

$922111 \rightarrow \frac{6!}{2!3!} = 60$

$433111 \rightarrow \frac{6!}{3!2!} = 60$

$= 120 + 90 + 60 + 60 + 30 + 15$

$= 375$

74. B

Sol. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ k & 0 & 0 \\ k & m & 0 \end{bmatrix}$

Let $\begin{bmatrix} 0 & 0 & 0 \\ k & 0 & 0 \\ k & m & 0 \end{bmatrix} = B \quad \therefore A = I + B$

$B^2 = \begin{bmatrix} 0 & 0 & 0 \\ k & 0 & 0 \\ k & m & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ k & 0 & 0 \\ k & m & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ mk & 0 & 0 \end{bmatrix}$

$B^3 = 0, B^4 = 0$

$(I + B)^n = I^n + {}^nC_1B + {}^nC_2B^2 + \ldots\ldots {}^nC_nB^n$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ nk & 0 & 0 \\ nk & nm & 0 \end{bmatrix} + \frac{n(n-1)}{2}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ mk & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ nk & 1 & 0 \\ nk + \frac{(n)(n-1)}{2}mk & nm & 1 \end{bmatrix}$$

$\Rightarrow nk = 16$

$\Rightarrow nm = 24$

$\Rightarrow nk + \frac{(n)(n-1)}{2}mk = 184 \Rightarrow n = 8, k = 2, m = 3.$

75. A

Sol. $\text{L.H.L.} = \lim\limits_{x \to 6^-} \frac{15 - [2 - x]}{[x - 2k]} = \lim\limits_{h \to 0} \frac{15 - [2 - 6 + h]}{[6 - h - 2k]} = \frac{19}{5 - 2k}$

$\text{R.H.L.} = \lim\limits_{x \to 6^+} \frac{15 - [2 - x]}{[x - 2k]} = \lim\limits_{h \to 0} \frac{15 - [2 - 6 - h]}{[6 + h - 2k]} = \frac{20}{6 - 2k}$

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76. C

Sol. $A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$

$\therefore B = \begin{bmatrix} \cos 2\theta + \cos 6\theta & \sin 2\theta + \sin 6\theta \\ -\sin 2\theta - \sin 6\theta & \cos 2\theta + \cos 6\theta \end{bmatrix}$

at $\theta = \frac{\pi}{8}$ $B = \begin{bmatrix} 0 & \sqrt{2} \\ -\sqrt{2} & 0 \end{bmatrix}$

77. A

Sol. $\begin{vmatrix} 1 & -2 & 3 \\ 0 & m & 1 \\ 3 & -24 & m \end{vmatrix} = 1(m^2 + 24) + 3(-2 - 3m)$

$\Rightarrow \quad m^2 - 9m + 18 \neq 0 \Rightarrow m \neq 3, 6 \ \therefore p = 2/3$

78. C

Sol. $\cot^{-1}\left(\sqrt{3}\right) + \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$

79. A

Sol. $f(0) = e^{\sqrt{3}/2}$

$f'(x) + f(x)\sqrt{1 - (\log f(x))^2} = 0$

$\Rightarrow \quad \frac{dy}{dx} = -y\sqrt{1 - (\ln y)^2} \quad \Rightarrow \int \frac{dy}{y\sqrt{1 - (\ln y)^2}} = -\int dx$

$\sin^{-1}(\ln y) = -x + c$

$\Rightarrow \quad \sin^{-1}((\ln f(x))) = -x + c$

at $x = 0$ $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = c \quad \Rightarrow c = \frac{\pi}{3}$

$\Rightarrow \quad \sin^{-1}(\ln f(x)) = -x + \frac{\pi}{3}$

$x = \frac{\pi}{4} \Rightarrow \sin^{-1}(\ln f(x)) = \frac{\pi}{12}$

$\Rightarrow \quad \ln f\left(\frac{\pi}{4}\right) = \sin\frac{\pi}{12}$

80. D

Sol.

p	q	~ p	~ p ⇔ q	~ q	~ q → (~p ⇔ q)
T	T	F	F	F	T
T	F	F	T	T	T
F	T	T	T	F	T
F	F	T	F	T	F

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SECTION – B

81. 7

Sol. $\frac{9\pi-2\pi}{2}=\frac{7\pi}{2}$

82. 7

Sol. Let $x^4 = u = \frac{du}{dx} = 4x^3 \Rightarrow f(t) = \frac{1}{4}\int_0^{t^4}\left(\frac{u^3e^u}{\left(u^3+3u^2+6u+1\right)^2}\right)du$

$$\Rightarrow f(t)=\frac{1}{4}\int_0^{t^4}e^u\left(\frac{1}{\left(u^3+3u^2+6u+1\right)}-\frac{3u^2+6u+1}{\left(u^3+3u^2+6u+1\right)^2}\right)du$$

$$f(t)=\frac{1}{4}\frac{e^u}{\left(u^3+3u^2+6u+1\right)}\Bigg|_0^{t^4}=\frac{et^4}{4\left(t^{12}+3t^8+6t^4+1\right)}-\frac{1}{4}$$

$$\Rightarrow f(1)=\frac{e}{44}-\frac{1}{4}$$

$$f'(t)=\frac{e^{t^4}\left(t^{15}\right)}{\left(t^{12}+3t^8+6t^4+1\right)^2} \Rightarrow f'(1)=\frac{e}{121}$$

83. 9

Sol. $f'(x+y) = f'(x) + 2y$

Put $x = 0$

$f'(y) = -1+ 2y$

$\Rightarrow f(y) = y^2 - y + c$

$\because f(0) = 3 \ \therefore f(y) = y^2 - y + 3$

84. 1

Sol. $x + y + z = p$

$x + 2ay + z = p$

$ax + a^2y + z = p$

$D = 0 \Rightarrow a = 1, \frac{1}{2}$

for $a = 1 \ D_x = D_y = D_z = 0$

85. 1

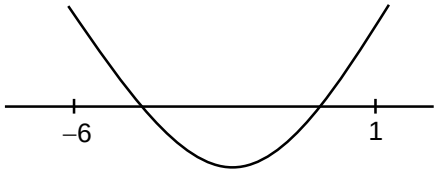
Sol. $$\frac{\sin\left(2^\circ+\left(\frac{89-1}{2}\right)2^\circ\right)\sin\left(\frac{89\times2^\circ}{2^\circ}\right)}{\sin\left(\frac{2^\circ}{2^\circ}\right)}$$

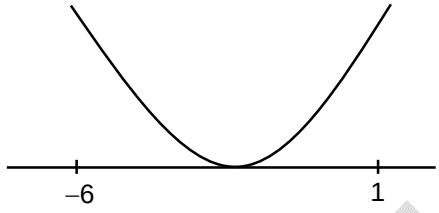
$$=\frac{\sin90^\circ\sin89^\circ}{\sin1^\circ}=\frac{\sin90^\circ\cos1^\circ}{\sin1^\circ}=\cot1^\circ \ \therefore k = 1.$$

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86. 6

Sol. $x^2 + 2(k-3)x + 9 = f(x)$





$\rightarrow D \geq 0 \Rightarrow k \in (-\infty, 0] \cup [6, \infty)$

$\rightarrow f(-6) > 0 \Rightarrow k < \frac{27}{4}$ $\quad f(1) > 0 \Rightarrow k > -2$

$\rightarrow -6 < -(k-3) < 1 \Rightarrow k \in (2, 9)$

$\Rightarrow k = 6$

$\Rightarrow 6 = 2 \times r^{21} \Rightarrow r^{21} = 3$

$a_4 a_{17} = 2r^4 \times 2r^{17} = 4r^{21} = 12$

87. 9

Sol. $\tan\theta \sec 2\theta = \frac{\sin\theta}{\cos\theta\cos 2\theta}$

$$\frac{\sin(2\theta-\theta)}{\cos\theta\cos 2\theta} = \tan 2\theta - \tan\theta$$

$$\sum(\tan 2\theta - \tan\theta) = \sum_{k=1}^{2025} \tan\left(\frac{\alpha}{2^{k-1}}\right) - \tan\left(\frac{\alpha}{2^k}\right)$$

$$= \tan\left(\frac{\alpha}{2^\circ}\right) - \tan\left(\frac{\alpha}{2^{2025}}\right)$$

88. 5

Sol. $\sum_{k=1}^{25} (k+1)^2 \, {}^{25}C_k - k^2 \, {}^{25}C_{k-1}$

$= 26^2 \, {}^{25}C_{25} - {}^{25}C_{25} = 676 - 1 = 675$

89. 3

Sol. $\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{\sqrt{3}}{4} \times 2 \Rightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = 3$

90. 9

Sol. $\frac{3^{140}}{4 \times 3} = \frac{3^{139}}{4} = \frac{(4-1)^{139}}{4}$

$(4-1)^{139} = {}^{139}C_0 4^{139} - {}^{139}C_1 \, 4^{138} + {}^{139}C_2 \, 4^{137} \ldots\ldots {}^{139}C_{138} \, 4 - {}^{139}C_{139}$

$= 4k - 1$

$\therefore$ remainder = 9

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